INCH-POUND

MIL-W-63377(AR)
AMENDMENT 6
6 January 1998
SUPERSEDING
AMENDMENT 5
10 October 1995

MILITARY SPECIFICATION

WARHEAD, ILLUMINATING, M257 W/M442 FUZE
(2.75 INCH ROCKET ILLUMINATING FLARE)

This amendment forms a part of Military Specification MIL-W-63377 (AR), dated 29 April 1987, and is approved for use by the US Army Armament, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1 Delete in its entirety and substitute:

"1.1 Scope. This specification covers the requirements, examinations and tests for the M257 Illuminating Flare and the M278 Infrared Flare, both components of the Hydra 70 rocket system. It covers the quality assurance provisions and special test requirements not covered by the drawings for parts and assembly of the M257 and M278 IR flare warheads (see 6.1).".

2.1.1: Add

"MIL-W-63468 - Warhead, 2.75 Inch Rocket, H.E.: M261".

PAGE 2

2.1.1: Add

"MIL-STD-1323-351 - Unit load for Underway Replenishment 2.75 Inch Warheads, M257, M261 or M267 in Container, PA 92".

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2.1.2: Delete

"8883478	-	Cont, Ammo, Fiber, M518, for rocket motor: 2.75 Inch, Mark 40 Mods, and Mark 4 and Mods
8883479	-	Box Packing Ammo for Rocket Motor, 2.75 Inch, Mark 40 and Mods, and Mark 4 and Mods
9347035	-	Packing and Marking for Box, Wood, Wirebound Pallet Type for Warhead 2.75 Inch Rocket Illuminating M257
9394492	-	Packing and Marking for Box, Wood Packing Ammunition for Warhead, 2.75 Inch Rocket Flare M257, Rocket Motor: 2.75 Inch Mark 40 and Mods, and Mark 4 and Mods
9340646	-	Box, Fiber, Packing, Ammunition for Warhead, 2.75 Inch Rocket Flare, M257
9340647	-	Pallet, (Boxed Type), for Warhead, 2.75 Inch Rocket: Illumination, M257".

2.1.2: Add

"9331197	-	Container, Ammo, Metal: PA 92
233AS43	-	Rocket Motor, Inert, MK66
1-2-601	-	Laboratory Inspection Schedules, International Test Operation Procedures (ITOP), TECOM
12944412	-	Warhead, 2.75 Inch Rocket Flare M278(IR)
12977529	-	Packing and Marking for Box, Wood Packing Ammunition for 2.75 Inch Warhead: M278 IR
1733AS300	-	Pack Shipping and Storage (For Four Warhead, 2.75 Inch Rocket Flare, M257)
3368AS200	-	Pack, Shipping and Storage (For 4 Warhead 2.75 Inch Rocket Flare M278 IR)".

Delete "19-48-4116/80M" and substitute "19-48-4116/80P".

3.2: Delete "on Drawing 9332220 and associated drawings" and substitute "on Drawing 9332220 (12944412 for the M278 IR Flare) and associated drawings".

3.4.1: Delete in its entirety and substitute:

"3.4.1 Function time. The time between rocket motor ignition and flare candle ignition shall be 13.5 ± 2.0 seconds with Mk40 motor and 13.5 ± 2.5 seconds with Mk66 motor."

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PAGE 4

3.4.2: Add

"After ignition, the M257 Warhead candle shall burn continuously (remain lit) during descent. There shall be no "chunking". Chunking is defined as a, broken off, lit section of candle that continues to burn on ground for an excess of 10 seconds. A chunk should not be confused with a burning warhead case."

3.4.3: Add

"3.4.3.1 M278 Burn time. The M278 IR Flare will provide infrared light for 180 seconds ± 30 seconds during flight testing, and burn time during tower testing collected for informational purposes only.".

3.5: Add

"3.5.1 M278 Illumination. The minimum output in the near infrared spectrum, defined as 0.7 to 1.1 microns wavelength, shall be 222 watts per steradian. The maximum light output in the visible spectrum, defined as 0.4 to 0.7 microns wavelength, shall be 1,000 candlepower with a desired output of 700 candlepower.".

PAGE 5

3.14, 2 places: Delete "Gas Generator" and substitute "2 second delay".

Delete "20 pounds is applied to the sear." and substitute "less than 20 ± .50 pounds is applied to the pull wire."

Add the following paragraph:

"3.14.1 Nonfunction Drop Test. The 2 second delay shall not function when subjected to seven foot drop to a steel plate five times in a pipe fixture."

3.15, 2 places: Delete "Gas Generator" and substitute "2 second delay".

3.17: Delete "50 pounds minimum and 75 pounds maximum" and substitute "40 pounds minimum and 65 pounds maximum".

3.18: Delete "16 g's plus or minus .5 g's" and substitute, "12 to 14 g's".

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3.19: Delete “twenty-two (22) g’s plus or minus .5 g’s” and substitute, “20 to 22 g’s”.

PAGE 7

4.3.1: Delete “Gas Generator 9332277 30” and substitute “2 Second Delay 9332277 50”.

Add:

“Post, Zig-Zag	9332274	10
Weight Assembly Setback, Zig-Zag	9332308	10
Weight, Setback, Zig-Zag	9332309	10
Warhead, 2.75 Inch Rocket Flare M278 (IR)	12944412	133
Candle and Parachute Assembly	12944413	10
Case Assembly, Loaded	12944415	50”.

PAGES 8-15

Table I, Under title: Delete “Warhead, 2.75 Inch Rocket Flare M257 and Components” and substitute “Warhead, 2.75 Inch Rocket Flare (Flare M257 or IR Flare M278) and Components”.

PAGE 11

Table I, Under “Gas Generator” Examination or Test: Add

“Nonfunction Drop Test	20	3.14.1	4.4.2.20”.

Delete “Gas Generator” and substitute “2 second delay”.

Delete “Case Assembly, Loaded (Dwg 9332239)” and substitute “Case Assembly, Loaded (Dwg 9332239/12944415 for M278)”.

Delete “Candle and Parachute Assembly (Dwg 9332222)” and substitute “Candle and Parachute Assembly (Dwg 9332222/12944413 for M278)”.

PAGE 14

Table I: Delete “Warhead, 2.75 Inch Rocket Flare M257 (Dwg 9332220)” and substitute “Warhead, 2.75 Inch Rocket (Flare M257 or IR Flare M278) (Dwg 9332220/12944412 for M278)”.

Delete:

"Forty-foot drop	6	3.10	4.5.5"

and substitute

"Forty-foot (12 meter) drop	6 (d)	3.10	4.5.5".

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Add:

“Weight Assembly Setback, Zig-Zag (Dwg 9332308) Examination for defects	10	3.2	4.4.2.38
Post, Zig-Zag (Dwg 9332274) Examination for defects	10	3.2	4.4.2.39
Weight, Setback, Zig-Zag (Dwg 9332308) Examination for defects	10	3.2	4.4.2.40”.

PAGE 15

Table I, Under Notes block, Add:

“(d) Use a sample of 8 warheads when testing with the PA 92 container.”

PAGE 16

4.4.1, Para. C: Delete “Magnesium” and substitute “Magnesium (Potassium Nitrate for M278)”.

PAGE 27

4.4.2.9: Delete Majors 101, 102, 103 and 104 in their entirety.

Delete Minors 201 and 202 in their entirety.

PAGE 34

4.4.2.16: Delete Major 102 in its entirety.

PAGE 38

4.4.2.20: Delete “Gas Generator” and substitute “2 second delay”.

Add Critical 2 as follows:

“2 Nonfunction Drop Test	10	---	3.14.1	4.5.7”.

PAGE 39

4.4.2.21, Under Drawing Number: Add “/12944415 for M278”.

Under Next Higher Assembly: Add “/12944414 for M278”.

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4.4.2.22, Under Drawing Number: Add “/12944413 for M278”.

Under Next Higher Assembly: Add “/12944412 for M278”.

PAGE 51

4.4.2.33: Delete Major 102 in its entirety.

PAGE 53

4.4.2.35, 2 places: Delete “Gas Generator” and substitute “2 second delay”.

Delete Major 108 in its entirety.

PAGE 55

4.4.2.37, Delete:

“Warhead, 2.75 Inch Rocket Flare M257” and substitute “Warhead, 2.75 Inch Rocket (Flare M257 or IR Flare M278)”.

Delete “Dwg 9332220” and substitute “Dwg 9332220/12944412 for M278”.

Add 4.4.2.38, included as PAGE 55a.

Add 4.4.2.39, included as PAGE 55b.

Add 4.4.2.40, included as PAGE 55c.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.38	Weight Assembly, Setback-Zig-Zag	1 OF 1	9332308

NEXT HIGHER ASSEMBLY: 9332273

CLASSIFICATION	EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.		
Major			
101	Pin not staked properly	3.2	Visual
Minor			
201	Evidence of poor workmanship	3.22	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.39	Post, Zig-Zag	1 OF 1	9332274

NEXT HIGHER ASSEMBLY: 9332273

CLASSIFICATION	EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.		
Major			
101	Post diameter, top	3.2	Gage
102	Post diameter, bottom	3.2	Gage
103	Post length	3.2	Gage
Minor			
201	Evidence of poor workmanship	3.22	Visual

NOTES:

AMSMC FORM 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.40	Weight, Setback, Zig-Zag	1 OF 1	9332309

NEXT HIGHER ASSEMBLY: 9332308

CLASSIFICATION	EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.		
Major	None defined.		
Minor			
201	Weight diameter	3.2	Gage
202	Weight length	3.2	Gage
203	Evidence of poor workmanship	3.22	Visual

NOTES:

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PAGE 58

4.5.3, Delete “except that dummy 2.75 Inch Rocket Motors” and substitute “except that inert 2.75 Inch Rocket Motors”.

Delete “in accordance with Method 514.2-7 of MIL-STD-810C. Resonance testing does not apply.” and substitute “in accordance with TECOM ITOP 1-2-601 (dtd. 1 Mar 88).”.

PAGE 59

Delete “performed in accordance with Method 514.2, Category d.3, procedure IIc, Table 514.2-IVA and figures 514.2-4B, 4c, 5D, 4E and 4F of MIL-STD-810C” and substitute

“performed in accordance with the vibration schedule listed in para. 4.5.4 of MIL-W-63468(AR), Warhead, 2.75 Inch Rocket, H.E.: M261”.

In second paragraph: Delete “The sample warheads, assembled with dummy 2.75 Inch Rocket Motors, shall be loaded in a Government furnished launcher.” and substitute

“The sample warheads, assembled with inert MK40 2.75 Inch Rocket Motors, shall be loaded in a Government approved launcher (M200 or M261). When sample warheads are assembled with inert MK66 2.75 Inch Rocket Motors, they shall be loaded in a Government approved launcher (M261).”.

PAGE 60

4.5.5: Delete in its entirety and substitute the following:

"4.5.5 Warhead forty-foot (12 meter) drop test. The forty-foot (12 meter) drop test, as specified in MIL-STD-331, shall be performed on either (6) six warheads (2 boxes) or (8) eight warhead (2 PA 92 containers) for First Article only. Each box/container shall be subjected to a forty-foot drop test.

a. For Army use, the warhead samples shall be packed and packaged in accordance with DWG. 12624792. Inert 2.75 inch rocket motors shall be used.

b. For Navy use, the warhead samples shall be packed and packaged in accordance with DWG. 1733AS300. No rocket motor required.

c. For either use, the orientations of the boxes/containers on striking shall be:

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1. One box or container - warheads nose down (longitudinal axis 45^{o} from vertical).

2. One box or container - warheads base down (longitudinal axis 45^{o} from vertical).

No individual box/container shall be subjected to more than one drop. Prior to performing this drop test, the following is necessary:

a. Mark each warhead with an x-ray plane.

b. X-ray each warhead in two planes 90^{o} opposed. Verify all components are secure and no armed fuzes are evident.

c. Visually verify that the zig-zag weight is in the up and safe position.

d. Serialize each warhead.

e. Before dropping, each packaged warhead and box/container shall be visually examined for evidence of incorrect packaging.

After performing the drop test on all warheads, each round shall be x-rayed, utilizing the same plane reference established during x-ray prior to the drop test. A visual examination shall be performed. Any warhead found unsafe shall cause the first article sample to be rejected.".

PAGE 61

4.5.7, 5 places: Delete “Gas Generator” and substitute “2 second delay”.

Delete “20 pounds is applied to the sear.” and substitute “20 $\pm$ .50 pounds on the pull wire.”.

Delete “10 pounds on the sear.” and substitute “10 $\pm$ .25 pounds on the pull wire.”

Delete “sear movement or gas generator function.” and substitute “pull wire extraction or 2 second delay function.”.

Add the following paragraph:

“4.5.7.1, 2 Second delay nonfunction drop test. For First Article, twenty (20) each 2 second delays shall be subject to a drop test. Each 2 second delay shall be loosely installed into a 0.50 in. dia. x 6.00 in. long steel pipe and dropped five (5) times. The ends of the pipe shall be capped to prevent ejecting

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the 2 second delay during testing. The test fixture shall be dropped horizontal from a height of seven (7) feet min. to a steel plate. The First Article sample shall be rejected if one or more 2 second delay has functioned.

For production, a sample of ten (10) 2 second delays from each production lot shall be dropped (5) times in a manner as described above. If one or more 2 second delay has functioned, the production lot shall be rejected."

4.5.8, 4 places: Delete "Gas Generator" and substitute "2 second delay".

Delete "$30^{o} \pm 2^{o}$" and substitute "$30^{o} - 2^{o}$"

Delete last sentence and substitute:

"The lot shall be rejected if two or more time delay failures occur or one other defective class is found."

PAGE 62

4.5.10, Delete "16 g's plus or minus .50 g's" and substitute "12 to 14 g's".

Last sentence: Delete "classed defective." and substitute "classed defective and removed from the lot."

4.5.11, Delete "22 g's plus or minus .50 g's" and substitute "20 to 22 g's".

Last sentence: Delete "classed defective." and substitute "classed defective and removed from the lot."

4.5.12, Last sentence: Delete "classed defective." and substitute "classed defective and removed from the lot."

4.5.13, Delete "A complete round left out of the conditioning cabinet/container over five minutes prior to firing shall be placed at ambient temperature for not less than eight hours and then reconditioned to the applicable temperature." and substitute the following:

"A complete round left out of the conditioning cabinet/container over five minutes prior to firing shall be placed back in the conditioning cabinet/container for not less than twice the actual time it has been out of conditioning prior to launch."

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PAGE 63

4.5.13, Para. a: Delete

"This time shall be 13.0 ± 2.0 seconds or 13.5 ± 2.0 seconds depending on the type of motor used. (see 3.4.1)." and substitute the following:

"This time shall be 13.5 ± 2.0 seconds or 13.5 ± 2.5 seconds depending on the type of motor used. (see 3.4.1).".

Para. c: Delete first sentence in its entirety and substitute: "The M257 flare candle shall burn for a minimum of 100 seconds. The M278 flare candle shall burn for a minimum of 150 seconds.".

4.5.14: Delete in its entirety and substitute:

"4.5.14 Flare candle test. The flare candle test shall be performed at the contractor's facility. A minimum of forty (40) candles shall be selected for the first article quantity and subjected to an output test when static tested on a light measuring tower. In addition, from each production lot of flare candles produced, a minimum of forty (40) candles shall be selected for and subjected to an output test when static tested on a light measuring tower. The sample of candle assemblies shall be in accordance with Dwg 9332239 for M257 flares and Dwg 12944415 for M278 IR flares. The samples shall be less Parachute Assembly and Igniter Assembly.

4.5.14.1 Candle acceptance criteria. M257 flare candles shall be accepted or rejected in accordance with MIL-STD-414, normal inspection, variability unknown, standard deviation method, Single Specification Limit Form 1 AQL=6.5% (Table B-1). The M278 IR flare shall use an AQL (Table B-1) of 2.5% when testing both the IR output and the visible output.".

PAGE 64

5.: Delete in entirety and substitute the following:

"5. PACKAGING

"5.1 Preservation and Packing. The warheads (M257 and M278) shall be preserved and packed in accordance with Dwg. 9294519 for Army packaging. For the Navy, the M257 Warhead shall be preserved and packed in accordance with Dwg. 1733AS300 and the M278 Warhead shall be preserved and packed in accordance with

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Dwg. 3368AS200. For the Air Force, the M278 shall be preserved and packed in accordance with Dwg. 12977529.

5.2 Packing. When shipping warheads (M257 and M278) with MK66 Rocket Motors, packing shall be in accordance with 12624792. When shipping the M257 Warhead in metal containers, packing shall be in accordance Dwg. 1733AS300. When shipping the M278 Warhead in metal containers, packing shall be in accordance Dwg. 3368AS200. For the Air Force, the M278 shall be packed in accordance with Dwg. 12977529."

5.3 Marking. For Army packaging (M257 and M278), markings shall be in accordance with 9294519 and 12624792. For the Navy, the M257 Warhead shall be marked in accordance with Dwg. 1733AS300 and the M278 Warhead shall be marked in accordance with Dwg. 3368AS200. For the Air Force, the M278 shall be marked in accordance with Dwg. 12977529.

5.4 Unitization. The Army and Air Force packages shall be unitized in accordance with DWG. 1948-4116/80P. Navy containers shall be unitized in accordance with MIL-STD-1323-351.

5.5 Shipping. When warheads from more than one lot are shipped at the same time, each lot shall be kept separate and the division between lots clearly indicated to prevent the mixing of lots in transit."

PAGE 65

6.1: Delete in its entirety and substitute:

"6.1 Intended use. The warheads covered in this specification are the M257 Flare and the M278 IR Flare. Both warheads are intended for use as part of the Hydra 70 Rocket System. Both warhead are to be used in either Army packing and packaging or Navy containers.".

6.2: Delete in its entirety and substitute the following:

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this requirement.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

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d. Packaging requirements as applicable to either the M257 Illuminating Flare or M278 Infrared Flare, "Army, Air Force or Navy packaging" must be identified for use (i.e. box or container). Any additional packaging requirements, if other than specified in Section 5.

e. Applicable stock number.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.".

6.5: Add “Infrared”.

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6.6: Delete in its entirety.

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